

PakTravel AI System

Intelligent Logistics & Transport Optimization

Section 1: Logistics

Presented by *Member 1: Route Planning & Search Algorithms*

PART 1: INTELLIGENT ROUTING

Developing optimal paths for inter-city travel across Pakistan's major nodes.

- ✓ Graph-based mapping of 15 cities
- ✓ Dijkstra & A* Implementation
- ✓ Relay routing for complex legs



ALGORITHM EFFICIENCY

A* Search



15 Nodes

Uniform Cost (UCS)



45 Nodes

A Search exhibits 3x efficiency by leveraging heuristic straight-line distance estimates.*

Section 2: **Management**

Presented by Member 2: Legal Advisor & Bus Scheduling

PART 2: AI LEGAL ADVISOR



Rule Engine

15 codified laws using
Propositional Logic.



Inference

Resolution Refutation to prove
outcomes.



Auto-Fining

Calculated fine of 2,000 PKR
for violations.

PART 3: BUS SCHEDULING

Route ID	Time	Assigned Bus	Constraints
R1: Karachi-Hyd	08:00 AM	B1 (Large Coach)	Highway Optimized
R3: Sukkur-Multan	12:00 PM	B5	Ali's Relay Bus
R7: Islamabad-Lhr	08:00 AM	B2	No Overlap (R1)

OPTIMIZATION SUCCESS

100%

Constraint Compliance

16/16

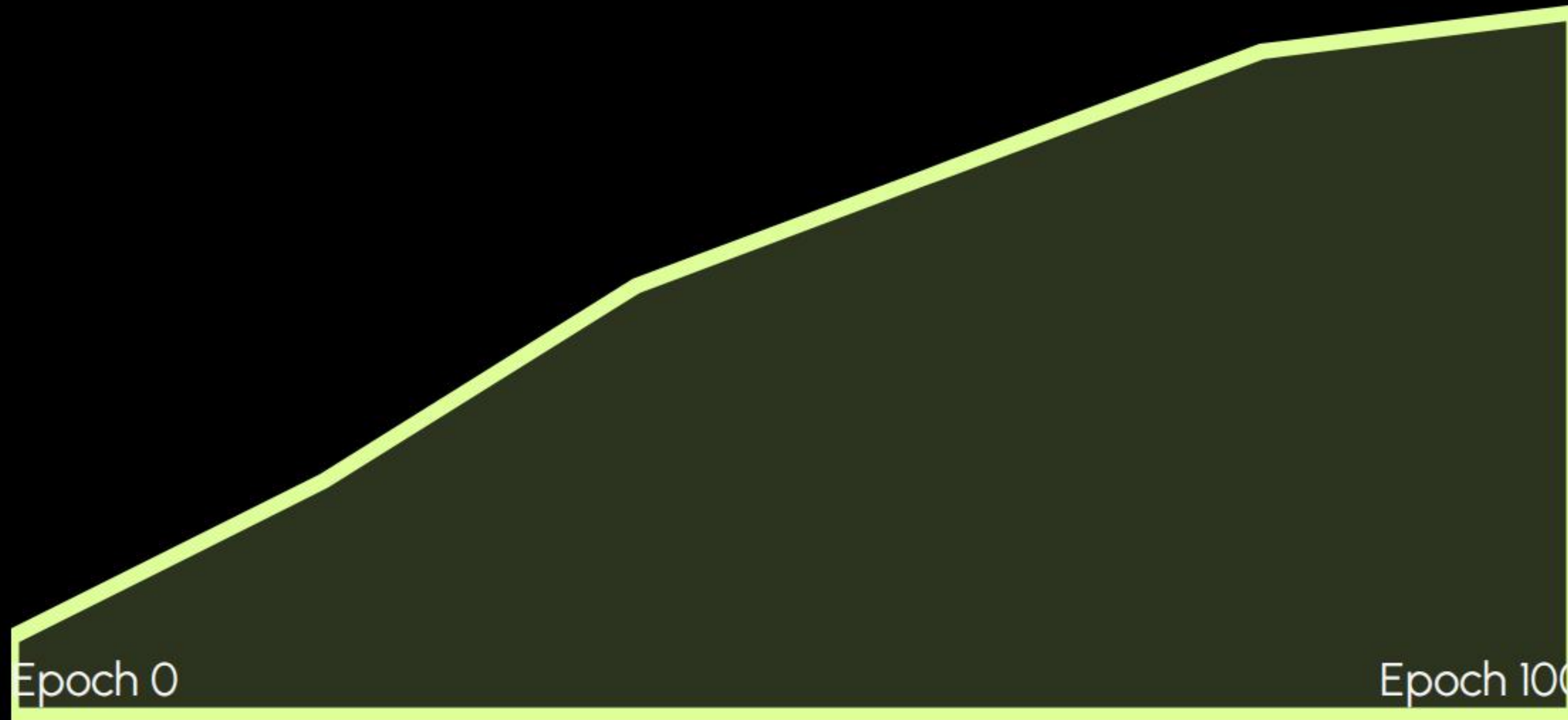
Optimization Score

Utilizing Backtracking with MRV Heuristic and AC-3 Arc Consistency.

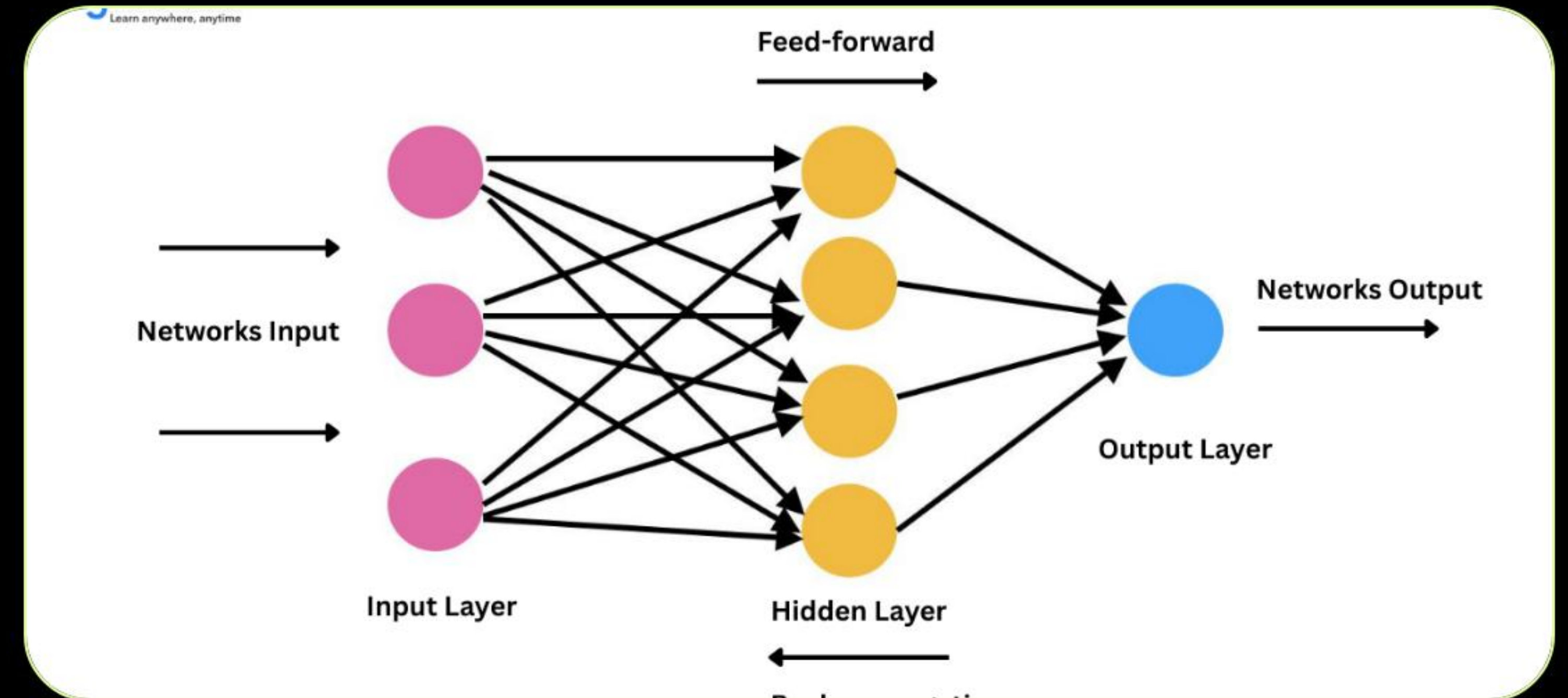
Section 3: Prediction

Presented by *Member 3: ANN Forecasting & Clustering*

PART 4: DELAY FORECASTING



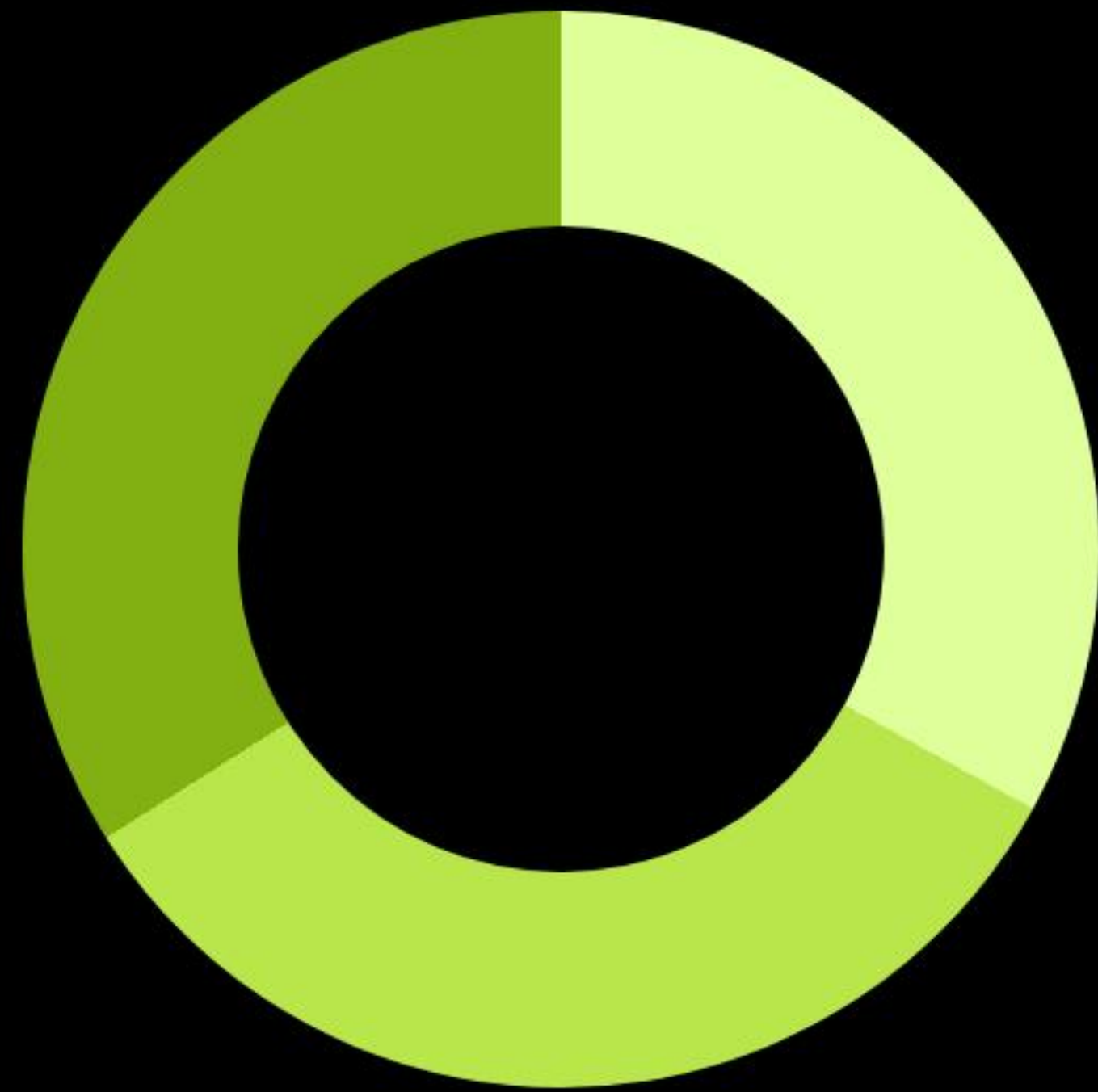
Training Loss Convergence Curve



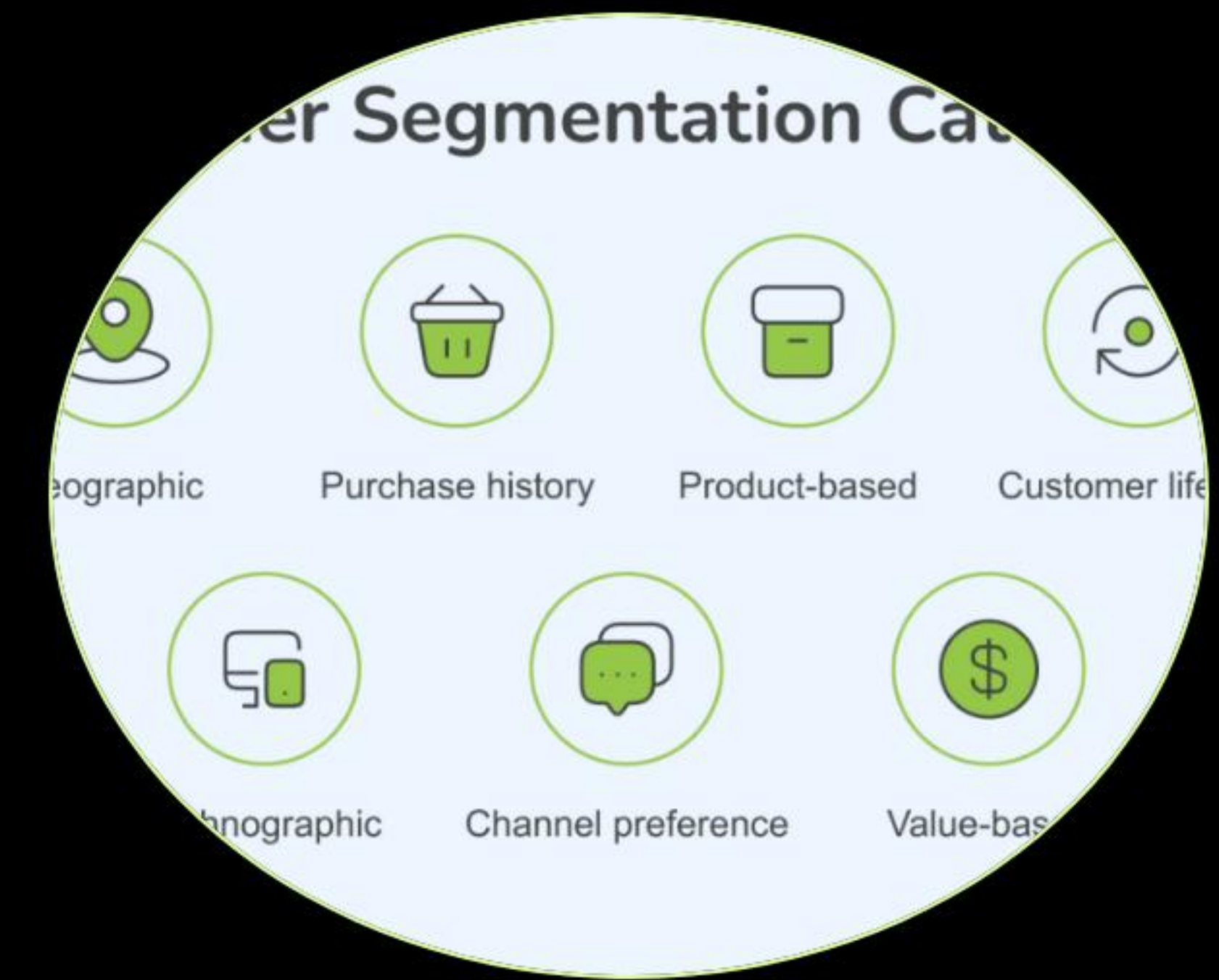
ANN Accuracy: 96%

Multi-Layer Perceptron (10, 8 Hidden Units) predicting real-time delays.

PART 5: CLUSTERING ANALYSIS



- Business Travellers (33%)
- Family Travellers (33%)
- Regular Commuters (34%)



The Future of Travel

PakTravel AI: Scaling Intelligence for a Connected Pakistan.

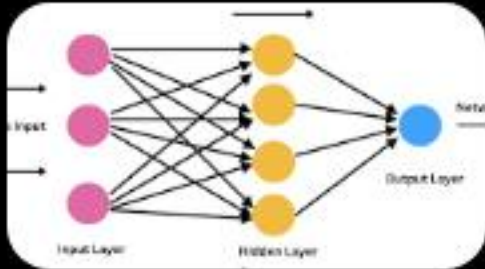
Q&A

IMAGE SOURCES



<https://static.vecteezy.com/system/resources/thumbnails/070/707/055/small/global-logistics-network-with-glowing-connections-over-world-map-and-shipping-ports-representing-international-trade-photo.jpeg>

Source: www.vecteezy.com



<https://cdn.shopaccino.com/igmguru/images/what-is-artificial-neural-network-3350062279421351.png>

Source: www.igmguru.com



https://m-img.org/spai/q_lossless+ret_img+to_webp/matomo.org/wp-content/uploads/2024/05/Customer-Segmentation-Categories-1024x571.png

Source: matomo.org
